BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 079 · Lesson 079 · Bubble Sort: Adjacent Swaps and Passes

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which elements does bubble sort compare?

Q2. What is placed by one ascending pass?

Q3. What does swap do?

Q4. Swap elements at indexes 0 and 1 in array [4,2,6]. Give the new array.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

bubble sort — Compare/swap adjacent values.

pass — Sweep through neighbouring pairs.

swap — Exchange two elements.

invariant — Always true state after given stage.

New learning

Understand — Compare adjacent pairs, swapping them when out of ascending order.

Trace — One completed ascending pass places the largest remaining element at the right end of the unsorted prefix.

Explain — Straightforward bubble sort is $O(n^2)$ worst time and O(1) extra memory; an early-exit flag can give O(n) best time.

Worked example

Show full first pass [4,1,3,2].

1. Check the working — [1,4,3,2]→[1,3,4,2]→[1,3,2,4].

We Do · Practise with guidance

Show full first pass [3,2,5,1].

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[2,3,5,1] (swap); 3,5 no swap; [2,3,1,5] (swap).
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Check the working

[2,3,5,1] (swap); 3,5 no swap; [2,3,1,5] (swap).

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Give all states from first pass [5,2,4,1]. [6]

Question 2

Give pass-end arrays for [5,2,4,1] until sorted. [6]

Question 3

Explain bubble-sort worst-time and extra-space complexity and benefit of early-stop. [6]

Exit Ticket · Answer before checking

After ascending first bubble pass, smallest value must be fixed at index0.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Adjacent pairs.

Q2. Largest unsorted value at end.

Q3. Exchanges two stored values.

Q4. [2,4,6].

You Do pages · Independent questions

Q1. [2,5,4,1] [2], [2,4,5,1] [2], [2,4,1,5] [2].

Q2. Pass1 [2,4,1,5] [2], pass2 [2,1,4,5] [2], pass3 [1,2,4,5] [2].

Q3. Worst $O(n^2)$ comparisons [2], O(1) in-place extra [2], no-swap pass can stop early [2].

Exit Ticket · Model answer

False. Largest is fixed at right end.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: